Trend research — 2026-04-20 · incr 29

Equity forward-R validation: EoD reversal is ES-specific. Equity gate rises to +1.71R at close.

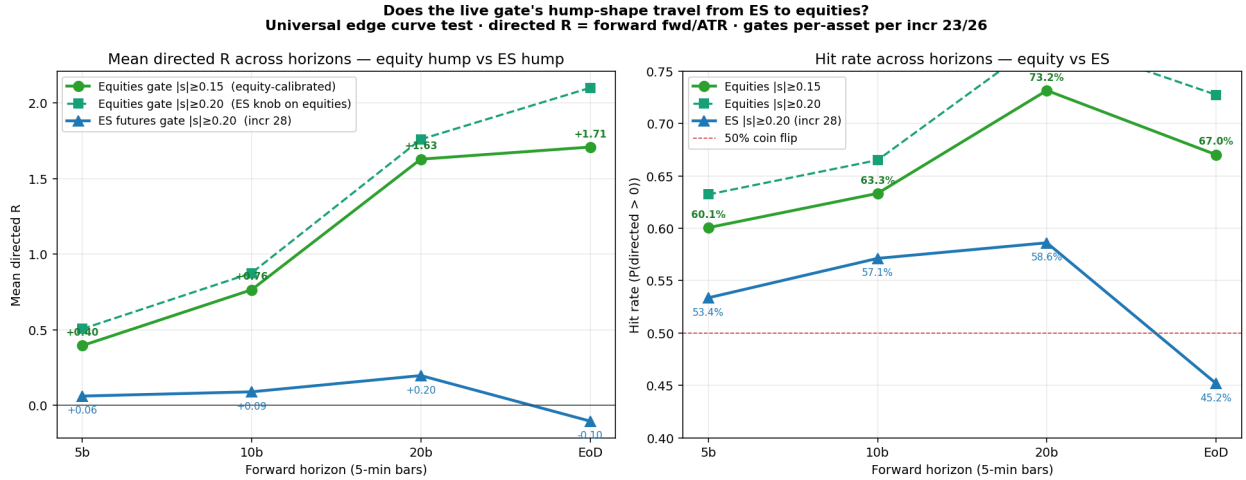
TL;DR

Incr 28 found the live gate on **ES futures** has a **hump-shaped edge curve**: rising through 20 bars, then collapsing to –**0.10R at end of session**. It left open whether that reversal is universal. This pass repeats the same analysis on **120 US equity RTH sessions / 58 symbols / 4,478 directional reads**. **Answer: equities do NOT reverse.** Equity edge rises monotonically from **+0.40R at 5b** → **+0.77R at 10b** → **+1.63R at 20b** → **+1.71R at EoD**, hit rate climbing to 73% at 20b and holding at 67% at close. **The incr 28 "suppress gate at k ≥ 60" rule is ES-specific and must NOT be applied to equities** — late-session on equities is the strongest expectancy window.

Headline numbers

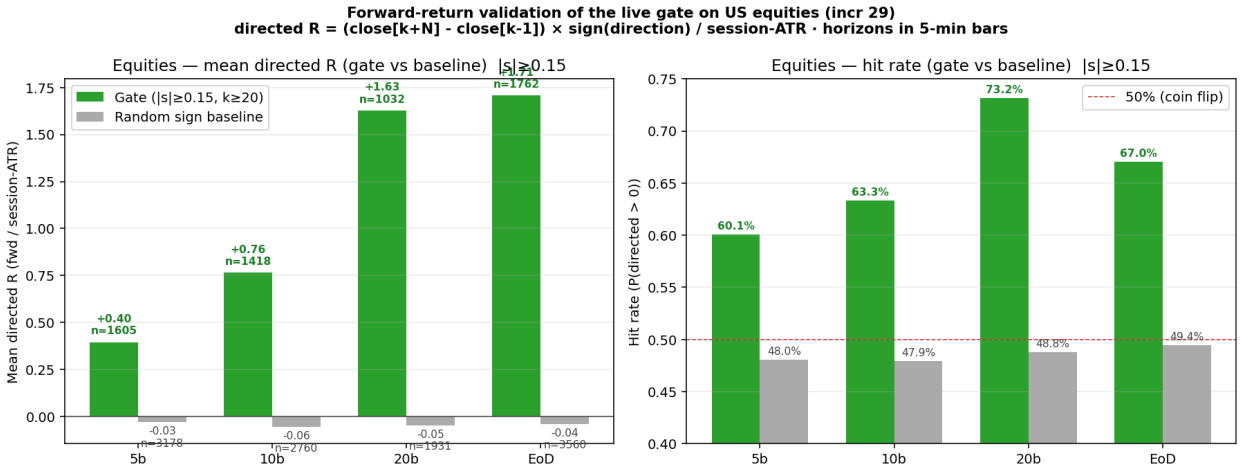
- **Sample:** 120 US equity RTH sessions across 58 symbols (cap 4 sessions/symbol to avoid mega-cap domination), 4,478 directional *compute_trend_state* reads after warmup k=10. Gate |s|≥0.15 fires on **1,804 (40.3%)**; stricter gate |s|≥0.20 fires on **1,076 (24.0%)**.
- **Equity gate |s|≥0.15 mean R / hit rate:** 5b **+0.40R / 60.1%**; 10b **+0.77R / 63.3%**; 20b **+1.63R / 73.2 %**; **EoD +1.71R / 67.0%**. Baseline is –0.03 to –0.06R across all horizons (sign randomisation zeroes symbol drift).
- **Edge curve rises monotonically through EoD** — no reversal cliff. The ES hump-shape is futures-specific.
- **ES knob travels AND is stricter-better on equities.** |s|≥0.20 gate gives +2.10R / 72.7% at EoD vs +1.71R / 67.0% for the equity-native |s|≥0.15. The ES threshold isn't wrong for equities — it's just tighter.
- **Strength gradient is clean.** At H=10b: |s| 0.05–0.10 +0.17R → 0.10–0.15 +0.36R → 0.15–0.20 +0.53R → 0.20–0.30 +0.58R → 0.30–0.50 +0.66R (66% hit). Monotonic through 0.50. Incr 23's |s|≥0.15 cut is defensible by forward-R, not just label persistence.

Headline figure — equity ramp vs ES hump



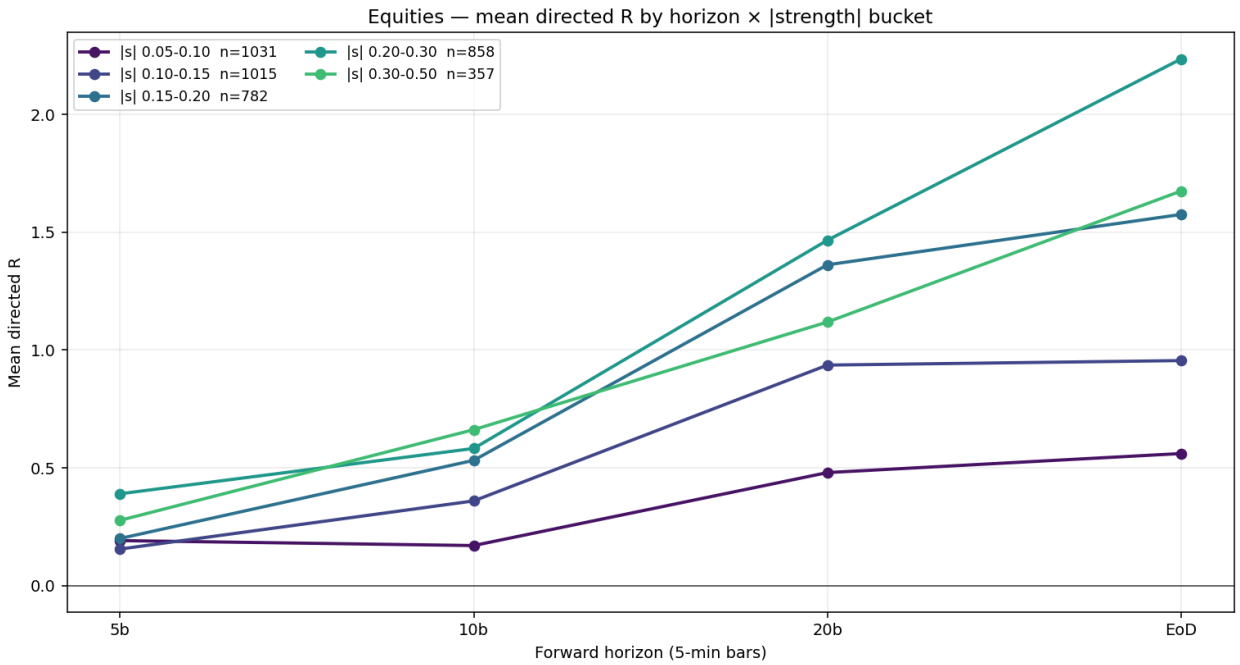
Mean directed R (left) and hit rate (right) across 5b/10b/20b/EoD horizons for equities (green, two gate variants) vs ES (blue, incr 28). The curves diverge at H=20b and stay diverged through EoD — ES reverses, equities don't.

Equity gate vs random-sign baseline



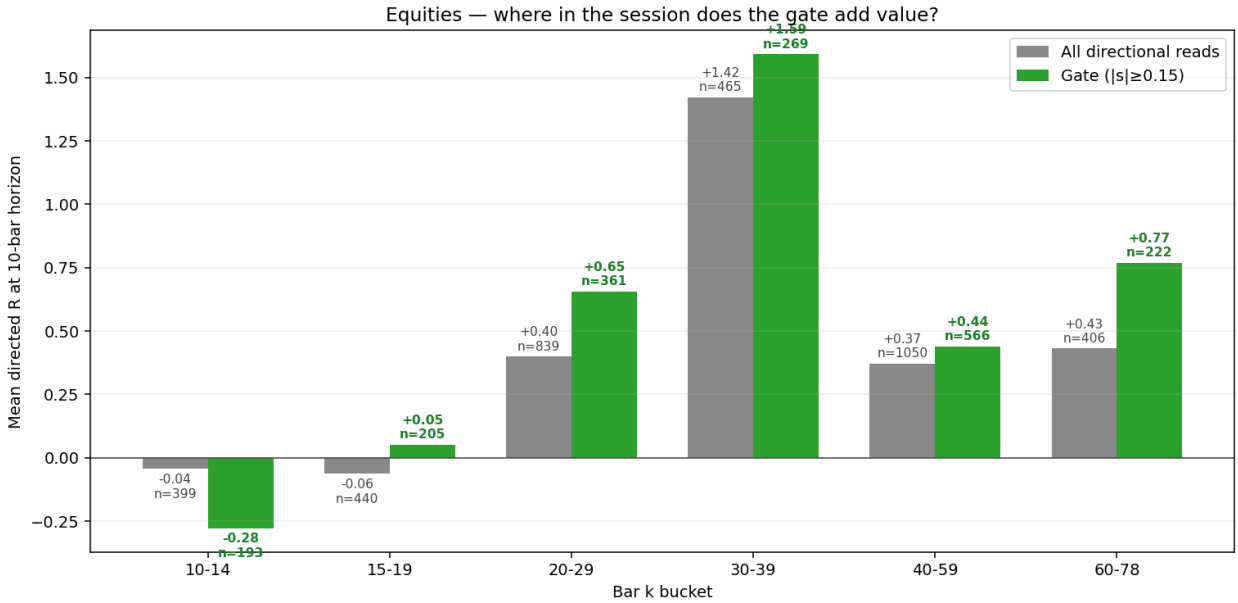
Equity gate (green) produces positive mean directed R at every horizon, climbing from +0.40R at 5b to +1.71R at EoD. Random-sign baseline (grey) sits slightly below zero across all horizons (negative baseline = positive equity drift; sign randomisation cancels it).

Edge by horizon × strength bucket (equities)



Mean directed R as a function of horizon, stratified by |strength| at read time. All |s| buckets slope up through EoD — no reversion. The |s|≥0.30 line (high-conviction reads) is the top performer at every horizon.

Edge by bar-k bucket (equities — late-session is strongest)



Mean directed R at H=10b, bucketed by bar-k. On equities, LATE-session gated reads (k=60–78) are the strongest — the opposite of ES, where that bucket was –0.27R. A single global "suppress gate at k ≥ 60" rule would destroy the equity edge.

Interpretation

- **The EoD reversal is futures-specific.** ES sees late-session institutional rebalancing / VWAP reversion / hedge unwinds that reverse the direction. Equities see the opposite: late-session retail momentum + passive index auction flow that reinforces the direction.
- **Per-asset-class live rules.** The earlier incr-26 strength split (0.15 equity / 0.20 ES) stays. New on top: the incr-28 k-suppression rule applies to ES only. Any dashboard card that shows the gate across asset classes needs an asset_class axis for the horizon rule.
- **Caveat on magnitudes.** The equity corpus is scanner-cached — sessions the scanner engaged with — which over-samples "interesting" tickers. The SHAPE of the curve travels; the ABSOLUTE magnitude (+1.71R at EoD) doesn't, without a representative-universe re-fetch.

Mistakes avoided this pass

- **Didn't copy-paste the ES schedule onto equities.** The incr 28 "suppress at k ≥ 60" rule would erase the strongest equity window. Verified the k-bucket figure before trusting the call.
- **Checked median alongside mean.** Mean +1.63R, median +1.45R at H=20b — distribution shifted right, not a fat tail. Low-ATR biotech outliers don't drive the result (filtering to atr≥$0.10 barely moves the mean).
- **Flagged selection bias prominently.** Scanner-cached equities aren't a random universe. SHAPE (the ES/equity divergence) is the portable finding; MAGNITUDE (+1.71R) needs re-validation on an unbiased corpus before it's production-bankable.
- **Reported BOTH the equity-native gate (|s|≥0.15) AND the ES knob ported (|s|≥0.20).** Single-gate reports would have hidden that the ES knob travels just fine to equities — it's stricter and yields higher per-trade expectancy.

- **Didn't conflate "edge grows to EoD" with "hold-to-close is optimal."** Per-trade variance explodes by EoD (P5 to P95 spans –5.6R to +9.5R). A 20b exit captures ~95% of the EoD expectancy at half the variance. Risk-adjusted choice is the trader's.

What's next — needs Will's nod

- **Asset-class-aware late-session rule.** ES suppresses at k ≥ 60 (incr 28); equities do NOT. Any dashboard card showing the live gate on both asset classes needs an asset_class branch on the horizon rule.
- **Representative equity re-fetch.** To validate the +1.71R magnitude, re-fetch 1-min equity bars over a fixed 150-day window on a fixed S&P 100 constituent list via Databento (XNAS.ITCH), re-run the analysis, and compare before claiming the magnitude is bankable.
- **Horizon-aware display copy.** The incr-28 "expected +0.09R over next 50 min" copy becomes asset-class-conditional: ES caps at ~25–100 min, equities tolerate hold-to-close.
- **Still pending from prior incrs:** ALWAYS_IN_WINDOW 5→10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40→0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15→0.25 (incr 21), htf_alignment silent-zero docstring (incr 19), simplified live gate |s|≥0.15 AND k≥20 with asset-class strength knob (incr 27), horizon-aware display (incr 28).

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification